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Experiment-03

Aim : To deploy multi-service applications under Docker Compose.

This experiment maps to LO4.

LO4 : Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes, and Cloud Infrastructure tools.

Theory :

Docker Compose is a tool used for defining, configuring, and running **multi-container Docker applications**. It allows developers to manage multiple services of an application using a single configuration file called compose.yaml or docker-compose.yml.

In a multi-service application, different components such as a web server, application server, database, and cache run in separate containers. Docker Compose simplifies the process of creating, networking, and managing these containers together.

A Compose file is written in **YAML format** and contains the details of each service, including:

- Docker images to use
- Ports to expose
- Environment variables
- Dependencies between services
- Networks and storage configurations

In this experiment, a multi-service application is deployed using four services:

1. **Nginx Web Server**
   - Acts as the web service and handles HTTP requests.
   - Provides communication between users and the application.
2. **Flask Application**

- Acts as the application layer.
- Runs the Python-based web application inside a Docker container.

3. **MySQL Database** ○ Stores and manages application data.
   - Runs as an independent database container.
4. **Redis Cache** ○ Provides fast in-memory data storage.
   - Improves application performance by caching frequently accessed data.

Docker Compose creates a separate container for each service and connects them through a default Docker network, allowing the containers to communicate with each other. The command: docker compose up -d --build

builds required images, creates containers, establishes networking, and starts all services in the background. The command:

docker compose down stops and removes the containers and network

created by Docker Compose.

Advantages of Docker Compose:

- Simplifies deployment of multi-container applications.
- Provides consistent application environments.
- Allows easy scaling of services.
- Reduces manual container management.
- Enables communication between related services through Docker networking.

Compose.yaml :

```
services:   web:
  image: nginx:latest
```

```
  ports:      -
"8081:80"
depends_on:
-           app
app:    build:
./app    ports:
- "5000:5000"
depends_on:
-           database
- cache
database:
image: mysql:8
environment:
    MYSQL_ROOT_PASSWORD: root
MYSQL_DATABASE: testdb   cache:
  image: redis:latest
```

Command	**Purpose**
docker --version	Checks the installed Docker version to verify Docker is available.
docker compose version	Checks the installed Docker Compose version.
mkdir dockercompose-demo	Creates a new project folder named docker-compose-demo.

Command	Purpose
cd docker-composedemo	Moves into the project folder where Docker Compose files are stored.
dir	Displays the files and folders inside the current directory.
notepad app.py	Creates/opens the app.py file using Notepad to write the Flask application code.
type app.py	Displays the contents of the app.py file in Command Prompt.
docker compose	Creates and starts the services defined in compose.yaml in detached mode (runs in the background).
docker compose --build	Builds Docker images again and starts all services in the background. Used when application code or Dockerfile changes.
docker ps	Shows currently running Docker containers.
docker ps -a	Shows all containers, including stopped and exited containers.
docker compose down	Stops and removes the containers, network, and resources created by Docker Compose.
docker compose	Displays logs from all services running through Docker Compose.
docker compose	Shows the status of services managed by Docker Compose.
docker info	Displays detailed information about the Docker installation and Docker Engine status.
docker version	Displays Docker client and server version information.

Output :

```
C:\Users\15L>docker --version
Docker version 29.2.1, build a5c7197

C:\Users\15L>docker compose version
Docker Compose version v5.0.2

C:\Users\15L>mkdir docker-compose-demo

C:\Users\15L>cd docker-compose-demo

C:\Users\15L\docker-compose-demo>docker compose up -d
unable to get image 'nginx:latest': error during connect: Get "http://%2F%2F.%2Fpipe%2FdockerDesktopLinuxEngine
he file specified.

C:\Users\15L\docker-compose-demo>docker compose up -d
[+] up 2/2
 ✔Network docker-compose-demo_default Created                                    0.0s
 ✔Container docker-compose-demo-web-1 Created                                    0.1s
Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:8080 -> 127.0.0.1:0: listen tcp
lly permitted.

C:\Users\15L\docker-compose-demo>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\15L\docker-compose-demo>docker compose down
[+] down 2/2
 ✔Container docker-compose-demo-web-1 Removed                                    0.0s
 ✔Network docker-compose-demo_default Removed                                    0.2s

C:\Users\15L\docker-compose-demo>docker compose up -d
[+] up 2/2
 ✔Network docker-compose-demo_default Created                                    0.0s
 ✔Container docker-compose-demo-web-1 Created                                    0.1s
```

```
C:\Users\15L\docker-compose-demo>docker compose up -d --build
#1 [internal] load local bake definitions
#1 reading from stdin 545B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 170B 0.0s done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.10
#3 DONE 4.5s

#4 [internal] load .dockerignore
#4 transferring context: 2B done
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 255B done
#5 DONE 0.0s

#6 [1/4] FROM docker.io/library/python:3.10@sha256:c4015e4e509b1aae50f742d32798ab65b8d08863305f61da5a0b0072d0070348
#6 resolve docker.io/library/python:3.10@sha256:c4015e4e509b1aae50f742d32798ab65b8d08863305f61da5a0b0072d0070348 0.0s done
#6 sha256:294bfa7c32fb5ee8d50148a1400cdc101b638116fceb05b1144036dd8e9f089c 0B / 249B 0.2s
#6 sha256:294bfa7c32fb5ee8d50148a1400cdc101b638116fceb05b1144036dd8e9f089c 249B / 249B 0.2s done
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 0B / 21.21MB 0.3s
#6 sha256:fd07ff3a0302154fe049b0332f7811aeaf927f44019e62cc6f97c2bcdc89b2a1 0B / 6.09MB 0.2s
#6 sha256:65956eab0c1ca261c14a3e13055ec8ca4f6ec75198f6ef0704d5194884466582 0B / 236.26MB 0.2s
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 2.10MB / 21.21MB 0.5s
#6 sha256:9e002cae30c445bfdf70841ba6a8d772e913914a00fa0e0f7bd0e4c089aaa838 0B / 67.79MB 0.2s
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 4.19MB / 21.21MB 0.6s
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 7.34MB / 21.21MB 0.8s
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 10.49MB / 21.21MB 0.9s
#6 sha256:fd07ff3a0302154fe049b0332f7811aeaf927f44019e62cc6f97c2bcdc89b2a1 1.05MB / 6.09MB 0.8s
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 12.58MB / 21.21MB 1.1s
#6 sha256:fd07ff3a0302154fe049b0332f7811aeaf927f44019e62cc6f97c2bcdc89b2a1 2.10MB / 6.09MB 0.9s
#6 sha256:fd07ff3a0302154fe049b0332f7811aeaf927f44019e62cc6f97c2bcdc89b2a1 4.19MB / 6.09MB 1.1s
#6 sha256:9e002cae30c445bfdf70841ba6a8d772e913914a00fa0e0f7bd0e4c089aaa838 4.19MB / 67.79MB 0.9s
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 15.73MB / 21.21MB 1.4s
#6 sha256:fd07ff3a0302154fe049b0332f7811aeaf927f44019e62cc6f97c2bcdc89b2a1 6.09MB / 6.09MB 1.3s done
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 19.92MB / 21.21MB 1.5s
#6 sha256:b453535073b6238aba838c5d063a1468b4d6a9aa54f81de64a4de304dc456685 0B / 25.64MB 0.2s
#6 sha256:77bce85bb12c2edfec770c62e757657111175d80ab22fb9373b701c91c5640078 21.21MB / 21.21MB 1.6s done
#6 sha256:9e002cae30c445bfdf70841ba6a8d772e913914a00fa0e0f7bd0e4c089aaa838 8.39MB / 67.79MB 1.4s
#6 sha256:b453535073b6238aba838c5d063a1468b4d6a9aa54f81de64a4de304dc456685 3.15MB / 25.64MB 0.5s
```

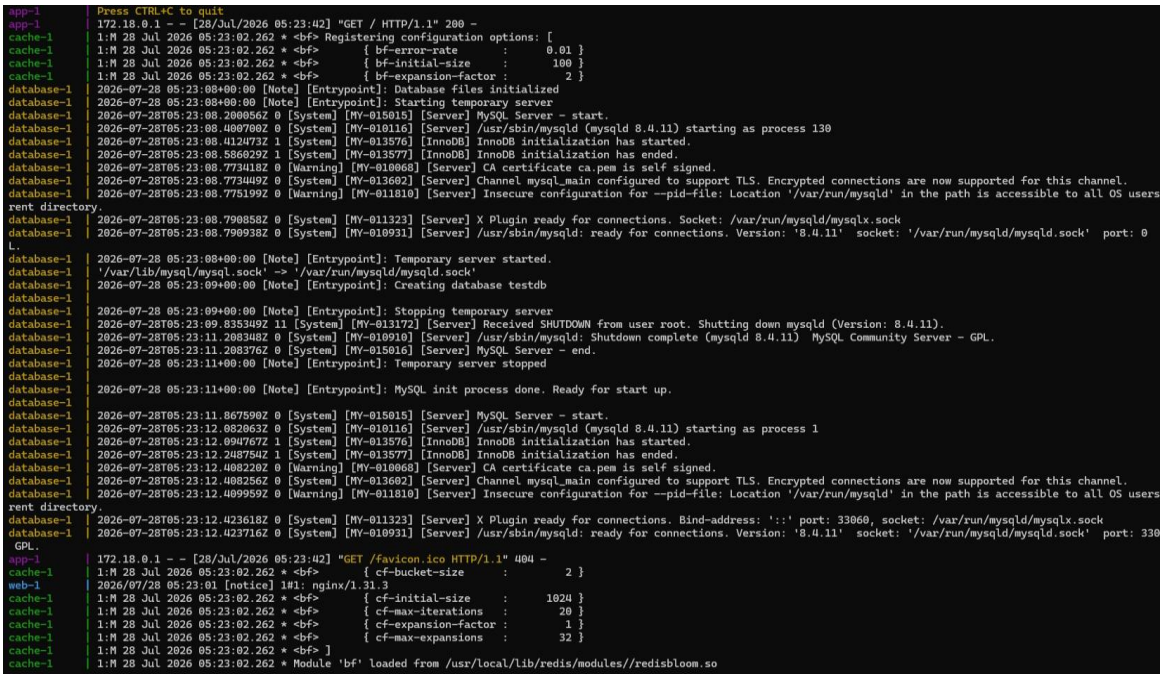
```
#9 2.708      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 21.7/21.7 MB 14.0 MB/s eta 0:00:00
#9 2.802 Collecting markupsafe>=2.1.1
#9 2.807   Downloading markupsafe-3.0.3-cp310-cp310-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
#9 2.833 Collecting click>=8.1.3
#9 2.837   Downloading click-8.4.2-py3-none-any.whl (119 kB)
#9 2.848      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 119.2/119.2 kB 12.0 MB/s eta 0:00:00
#9 2.861 Collecting blinker>=1.9.0
#9 2.865   Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
#9 2.880 Collecting itsdangerous>=2.2.0
#9 2.884   Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
#9 2.915 Collecting werkzeug>=3.1.0
#9 2.920   Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
#9 2.938      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 226.5/226.5 kB 13.7 MB/s eta 0:00:00
#9 2.957 Collecting jinja2>=3.1.2
#9 2.961   Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
#9 2.975      ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 kB 11.1 MB/s eta 0:00:00
#9 2.994 Collecting async-timeout>=4.0.3
#9 2.999   Downloading async_timeout-5.0.1-py3-none-any.whl (6.2 kB)
#9 3.051 Installing collected packages: mysql-connector-python, markupsafe, itsdangerous, click, blinker, async-timeout, werkzeug, redis, jinja2, f
#9 3.720 Successfully installed async-timeout-5.0.1 blinker-1.9.0 click-8.4.2 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 mysql-co
#9 3.720 WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is
pypa.io/warnings/venv
#9 3.776
#9 3.776 [notice] A new release of pip is available: 23.0.1 -> 26.1.2
#9 3.776 [notice] To update, run: pip install --upgrade pip
#9 DONE 3.9s

#10 exporting to image
#10 exporting layers
#10 exporting layers 2.1s done
#10 exporting manifest sha256:2fac7352405fc2dada7a25ed13f6911edf9f7ca4cdda57c41fedafefbe723e6f done
#10 exporting config sha256:d096278694acb7c94fd0f4530921a0ab263c1e712607fb951a100346f9cf283e done
#10 exporting attestation manifest sha256:e568f082cb2661f00a5f26b02112e3399bae4c9ccce6227a0bd21ed146a20eb9 0.0s done
#10 exporting manifest list sha256:96d8ab2f5b365a79035b40bae66fbf27b9e56efa94cda2d911a043d9c1ca6610 done
#10 naming to docker.io/library/docker-compose-demo-app:latest done
#10 unpacking to docker.io/library/docker-compose-demo-app:latest
#10 unpacking to docker.io/library/docker-compose-demo-app:latest 0.4s done
#10 DONE 2.6s

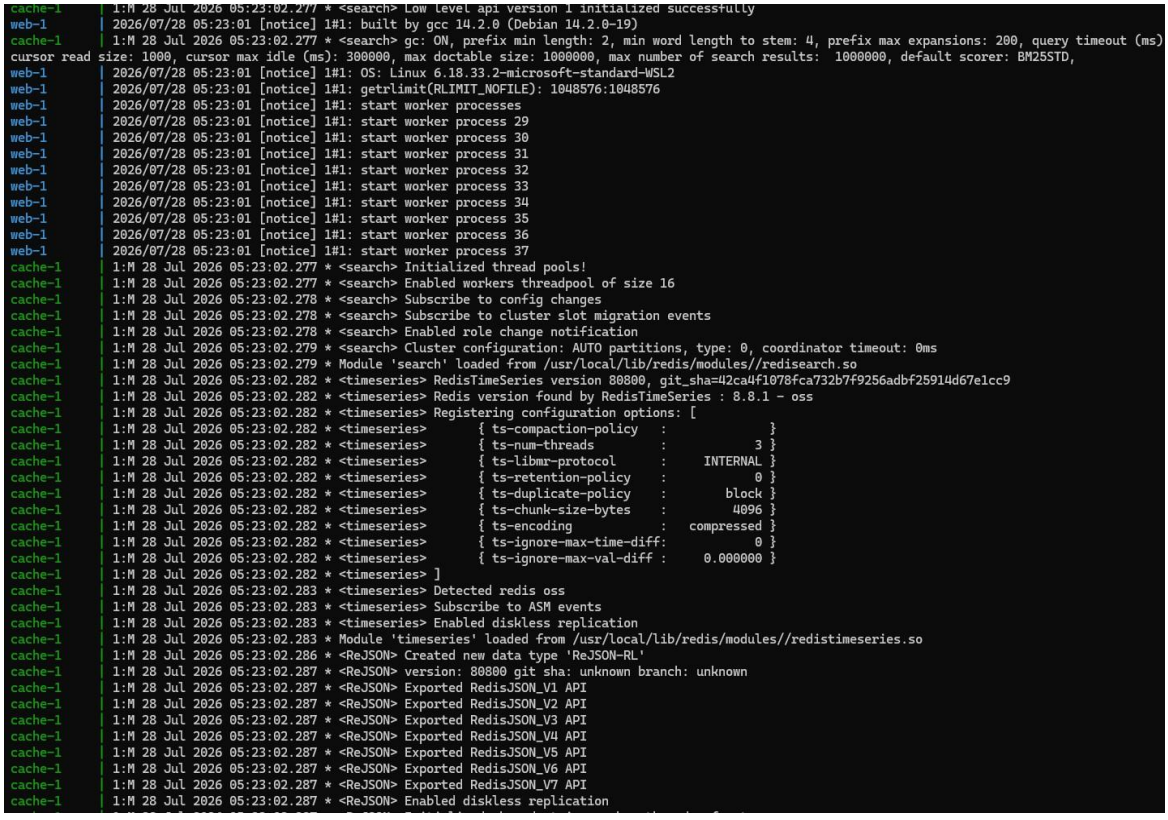
#11 resolving provenance for metadata file
#11 DONE 0.0s
[+] up 6/6
 ✔Image docker-compose-demo-app             Built                                    32.4s
 ✔Network docker-compose-demo_default       Created                                  0.0s
 ✔Container docker-compose-demo-database-1  Created                                  0.1s
 ✔Container docker-compose-demo-cache-1     Created                                  0.1s
 ✔Container docker-compose-demo-app-1       Created                                  0.0s
 ✔Container docker-compose-demo-web-1       Created                                  0.1s
```

```
C:\Users\15L\docker-compose-demo>docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS         PORTS                                                NAMES
a9354ba0d3fb   nginx:latest           "/docker-entrypoint.…"   7 seconds ago   Up 9 seconds   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp              docker-compose-demo-web-1
d3d3ad33eb83   docker-compose-demo-app   "python app.py"          7 seconds ago   Up 9 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp          docker-compose-demo-app-1
3804772d235c   redis:latest           "docker-entrypoint.s…"   7 seconds ago   Up 9 seconds   6379/tcp                                             docker-compose-demo-cache-1
62c184dfff88   mysql:8                "docker-entrypoint.s…"   7 seconds ago   Up 9 seconds   3306/tcp, 33060/tcp                                  docker-compose-demo-database-1

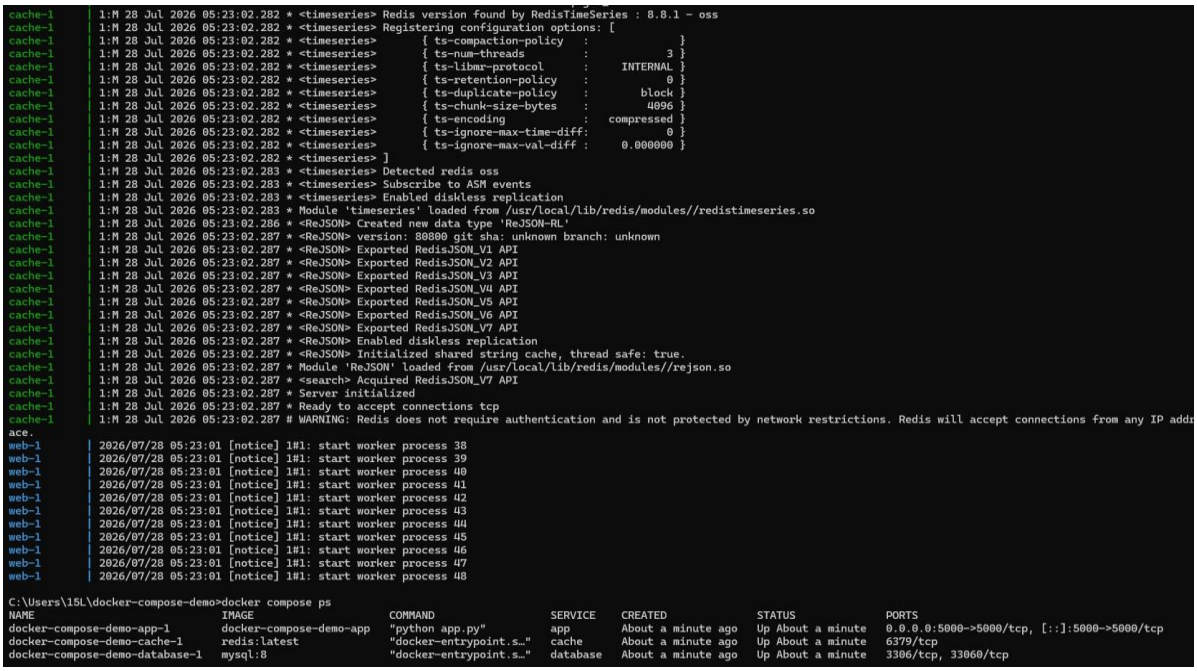
C:\Users\15L\docker-compose-demo>docker compose logs
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
database-1  | 2026-07-28 05:23:02+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.4.11-1.el9 started.
web-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
database-1  | 2026-07-28 05:23:01+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
database-1  | 2026-07-28 05:23:01+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.4.11-1.el9 started.
database-1  | 2026-07-28 05:23:05+00:00 [Note] [Entrypoint]: Initializing database files
database-1  | 2026-07-28T05:23:05.307231Z 0 [System] [MY-015017] [Server] MySQL Server Initialization - start.
cache-1     | Starting Redis Server
cache-1     | 1:C 28 Jul 2026 05:23:02.260 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
cache-1     | 1:C 28 Jul 2026 05:23:02.260 * Redis version=8.8.1, bits=64, commit=00000000, modified=1, pid=1, just started
database-1  | 2026-07-28T05:23:05.308007Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.4.11) initializing of server in progress as process 80
web-1       | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
app-1       |  * Serving Flask app 'app'
cache-1     | 1:C 28 Jul 2026 05:23:02.260 * Configuration loaded
database-1  | 2026-07-28T05:23:05.312142Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1  | 2026-07-28T05:23:05.456102Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
app-1       |  * Debug mode: off
cache-1     | 1:M 28 Jul 2026 05:23:02.260 * monotonic clock: POSIX clock_gettime
web-1       | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1       | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1       | /docker-entrypoint.sh: Configuration complete; ready for start up
cache-1     | 1:M 28 Jul 2026 05:23:02.261 * Running mode=standalone, port=6379.
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf> RedisBloom version 8.8.2 (Git=unknown)
app-1       | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
app-1       |  * Running on all addresses (0.0.0.0)
app-1       |  * Running on http://127.0.0.1:5000
app-1       |  * Running on http://172.18.0.4:5000
web-1       | 2026/07/28 05:23:01 [notice] 1#1: using the "epoll" event method
database-1  | 2026-07-28T05:23:06.121844Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure
database-1  | 2026-07-28T05:23:08.128770Z 0 [System] [MY-015018] [Server] MySQL Server Initialization - end.
app-1       | Press CTRL+C to quit
app-1       | 172.18.0.1 - - [28/Jul/2026 05:23:42] "GET / HTTP/1.1" 200 -
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf> Registering configuration options: [
```



```
app-1       | Press CTRL+C to quit
app-1       | 172.18.0.1 - - [28/Jul/2026 05:23:42] "GET / HTTP/1.1" 200 -
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf> Registering configuration options: [
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf>       { bf-error-rate      :      0.01 }
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf>       { bf-initial-size    :       100 }
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf>       { bf-expansion-factor :        2 }
database-1  | 2026-07-28 05:23:08+00:00 [Note] [Entrypoint]: Database files initialized
database-1  | 2026-07-28 05:23:08+00:00 [Note] [Entrypoint]: Starting temporary server
database-1  | 2026-07-28T05:23:08.200056Z 0 [System] [MY-015015] [Server] MySQL Server - start.
database-1  | 2026-07-28T05:23:08.400700Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.4.11) starting as process 130
database-1  | 2026-07-28T05:23:08.412473Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1  | 2026-07-28T05:23:08.586029Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
database-1  | 2026-07-28T05:23:08.773418Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
database-1  | 2026-07-28T05:23:08.773449Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
database-1  | 2026-07-28T05:23:08.775199Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users
rent directory.
database-1  | 2026-07-28T05:23:08.790858Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqlx.sock
database-1  | 2026-07-28T05:23:08.790938Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.4.11'  socket: '/var/run/mysqld/mysqld.sock'  port: 0
L.
database-1  | 2026-07-28 05:23:08+00:00 [Note] [Entrypoint]: Temporary server started.
database-1  | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
database-1  | 2026-07-28 05:23:09+00:00 [Note] [Entrypoint]: Creating database testdb
database-1  |
database-1  | 2026-07-28 05:23:09+00:00 [Note] [Entrypoint]: Stopping temporary server
database-1  | 2026-07-28T05:23:09.835349Z 11 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 8.4.11).
database-1  | 2026-07-28T05:23:11.208348Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.4.11)  MySQL Community Server - GPL.
database-1  | 2026-07-28T05:23:11.208376Z 0 [System] [MY-015016] [Server] MySQL Server - end.
database-1  | 2026-07-28 05:23:11+00:00 [Note] [Entrypoint]: Temporary server stopped
database-1  |
database-1  | 2026-07-28 05:23:11+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.
database-1  |
database-1  | 2026-07-28T05:23:11.867590Z 0 [System] [MY-015015] [Server] MySQL Server - start.
database-1  | 2026-07-28T05:23:12.082063Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.4.11) starting as process 1
database-1  | 2026-07-28T05:23:12.094767Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1  | 2026-07-28T05:23:12.248754Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
database-1  | 2026-07-28T05:23:12.408220Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
database-1  | 2026-07-28T05:23:12.408256Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
database-1  | 2026-07-28T05:23:12.409959Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users
rent directory.
database-1  | 2026-07-28T05:23:12.423618Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
database-1  | 2026-07-28T05:23:12.423716Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.4.11'  socket: '/var/run/mysqld/mysqld.sock'  port: 330
 GPL.
app-1       | 172.18.0.1 - - [28/Jul/2026 05:23:42] "GET /favicon.ico HTTP/1.1" 404 -
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf>       { cf-bucket-size     :         2 }
web-1       | 2026/07/28 05:23:01 [notice] 1#1: nginx/1.31.3
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf>       { cf-initial-size    :      1024 }
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf>       { cf-max-iterations  :        20 }
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf>       { cf-expansion-factor :        1 }
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf>       { cf-max-expansions  :        32 }
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * <bf> ]
cache-1     | 1:M 28 Jul 2026 05:23:02.262 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
```



```
cache-1     | 1:M 28 Jul 2026 05:23:02.277 * <search> Low level api version 1 initialized successfully
web-1       | 2026/07/28 05:23:01 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
cache-1     | 1:M 28 Jul 2026 05:23:02.277 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms)
cursor read size: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
web-1       | 2026/07/28 05:23:01 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
web-1       | 2026/07/28 05:23:01 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker processes
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 29
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 30
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 31
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 32
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 33
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 34
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 35
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 36
web-1       | 2026/07/28 05:23:01 [notice] 1#1: start worker process 37
cache-1     | 1:M 28 Jul 2026 05:23:02.277 * <search> Initialized thread pools!
cache-1     | 1:M 28 Jul 2026 05:23:02.277 * <search> Enabled workers threadpool of size 16
cache-1     | 1:M 28 Jul 2026 05:23:02.278 * <search> Subscribe to config changes
cache-1     | 1:M 28 Jul 2026 05:23:02.278 * <search> Subscribe to cluster slot migration events
cache-1     | 1:M 28 Jul 2026 05:23:02.278 * <search> Enabled role change notification
cache-1     | 1:M 28 Jul 2026 05:23:02.279 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
cache-1     | 1:M 28 Jul 2026 05:23:02.279 * Module 'search' loaded from /usr/local/lib/redis/modules//redisearch.so
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries> RedisTimeSeries version 80800, git_sha=42ca4f1078fca732b7f9256adbf25914d67e1cc9
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries> Redis version found by RedisTimeSeries : 8.8.1 - oss
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries> Registering configuration options: [
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-compaction-policy   :             }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-num-threads         :           3 }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-libmr-protocol      :    INTERNAL }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-retention-policy    :           0 }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-duplicate-policy    :       block }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-chunk-size-bytes    :        4096 }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-encoding            :  compressed }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-ignore-max-time-diff:           0 }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries>       { ts-ignore-max-val-diff :    0.000000 }
cache-1     | 1:M 28 Jul 2026 05:23:02.282 * <timeseries> ]
cache-1     | 1:M 28 Jul 2026 05:23:02.283 * <timeseries> Detected redis oss
cache-1     | 1:M 28 Jul 2026 05:23:02.283 * <timeseries> Subscribe to ASM events
cache-1     | 1:M 28 Jul 2026 05:23:02.283 * <timeseries> Enabled diskless replication
cache-1     | 1:M 28 Jul 2026 05:23:02.283 * Module 'timeseries' loaded from /usr/local/lib/redis/modules//redistimeseries.so
cache-1     | 1:M 28 Jul 2026 05:23:02.286 * <ReJSON> Created new data type 'ReJSON-RL'
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> version: 80800 git sha: unknown branch: unknown
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> Exported RedisJSON_V1 API
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> Exported RedisJSON_V2 API
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> Exported RedisJSON_V3 API
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> Exported RedisJSON_V4 API
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> Exported RedisJSON_V5 API
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> Exported RedisJSON_V6 API
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> Exported RedisJSON_V7 API
cache-1     | 1:M 28 Jul 2026 05:23:02.287 * <ReJSON> Enabled diskless replication
```





Multi-service Docker Compose Application Running!

Conclusion :

From this experiment, I understood that Docker Compose is used to deploy and manage **multi-service applications** by running multiple containers together. I learned how to create a compose.yaml file to define different services and how Docker Compose automatically creates, connects, and manages these containers.

I successfully deployed a 4-service application consisting of **Nginx, Flask application, MySQL database, and Redis cache**. I understood that each service runs independently in its own container but can communicate with other services through Docker networking.

I also learned how to build, start, check, and stop services using Docker Compose commands such as docker compose up, docker ps, and docker compose down. This experiment helped me understand how Docker Compose simplifies the deployment of complex applications with multiple components.